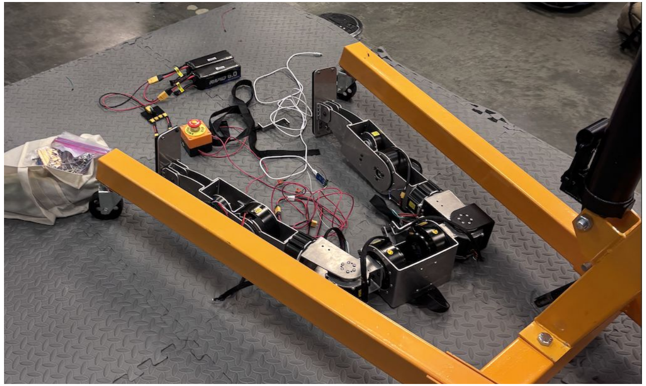
Humanoid Teleop

Embedded: Parth, Neal, Malik
ML: Darin, Cindy, Fong-Yu



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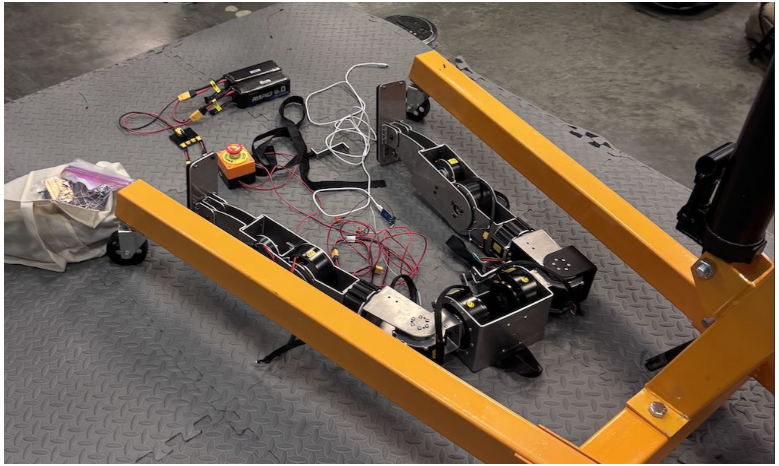
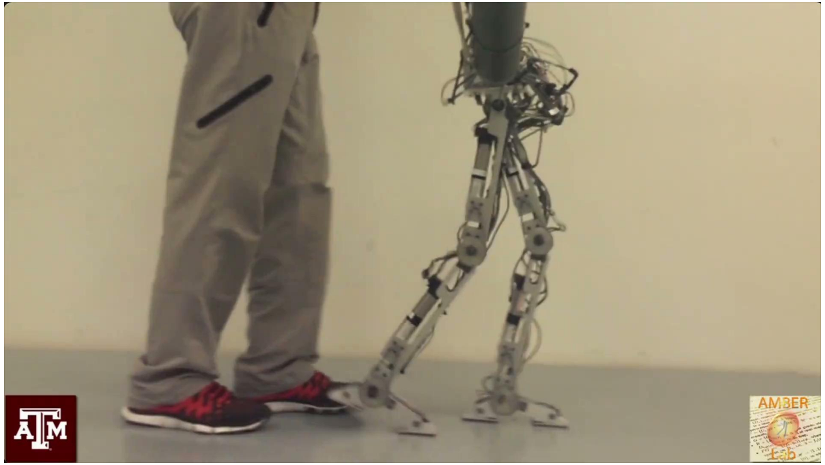
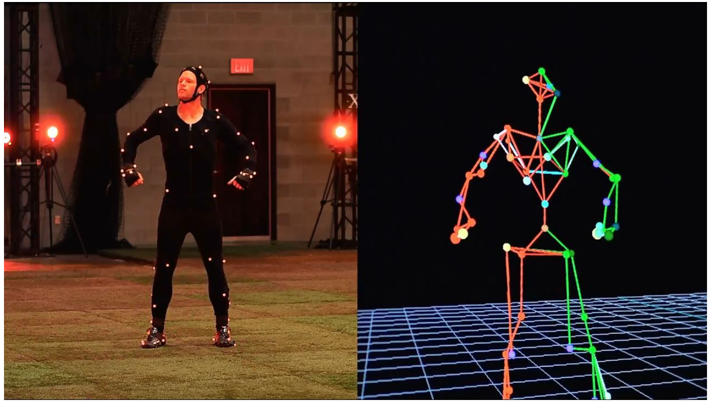
Overview: What are we doing?





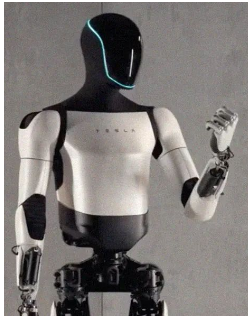
[Human-Like Multi-Contact Walking with AMBER 2]

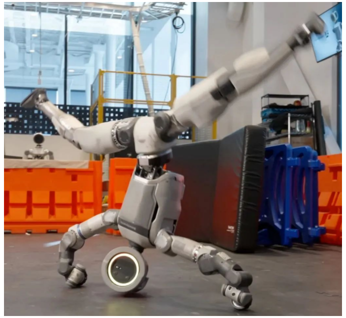
Overview: Problems We’re Solving



- People who are doing robot teleoperations are using very expensive equipments like mocap (~$3,000 cheapest) or VR devices costing north of $900 for full body gear.

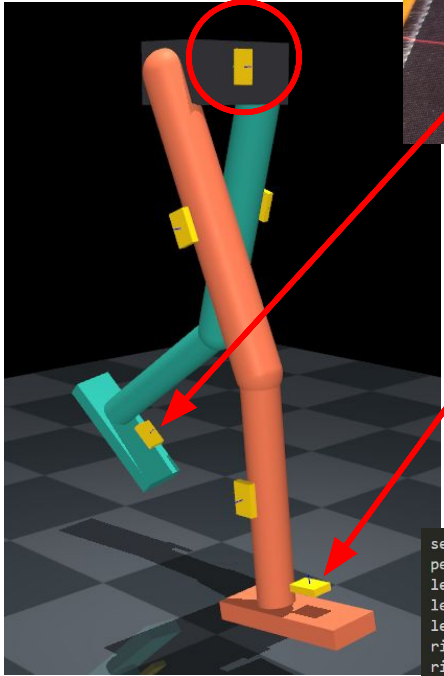
  However, we’re only using 7 ESP + 7 IMU's which costs ~$250 in total

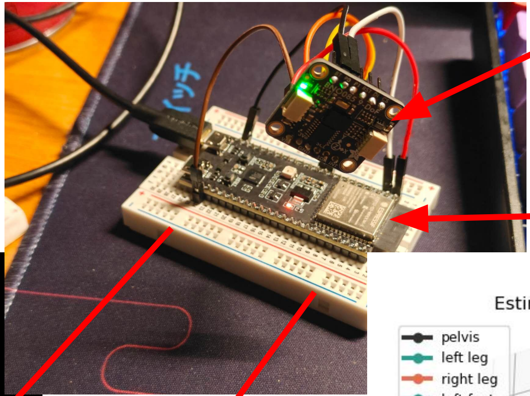




- Challenges:
  - Embedded devices connections…
  - Retargeting kinematic mismatch
    - Our legs proportions are different from the robot.
  - Noisy real-time input -> policy robustness
    - Existing papers such as DeepMimic or H2O use ground-truth motion to train the model. However, our input is real-time data which drifts and dropout (Sim2Real)

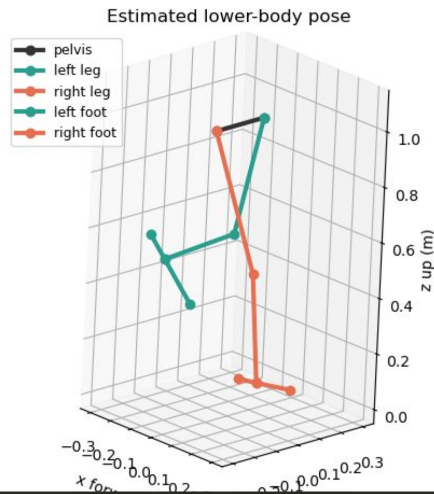
Overview





IMU: Tells the robot where it is facing and turning

ESP32: Reads the orientation of segments and send them wirelessly to central computing unit (Jetson Nano)

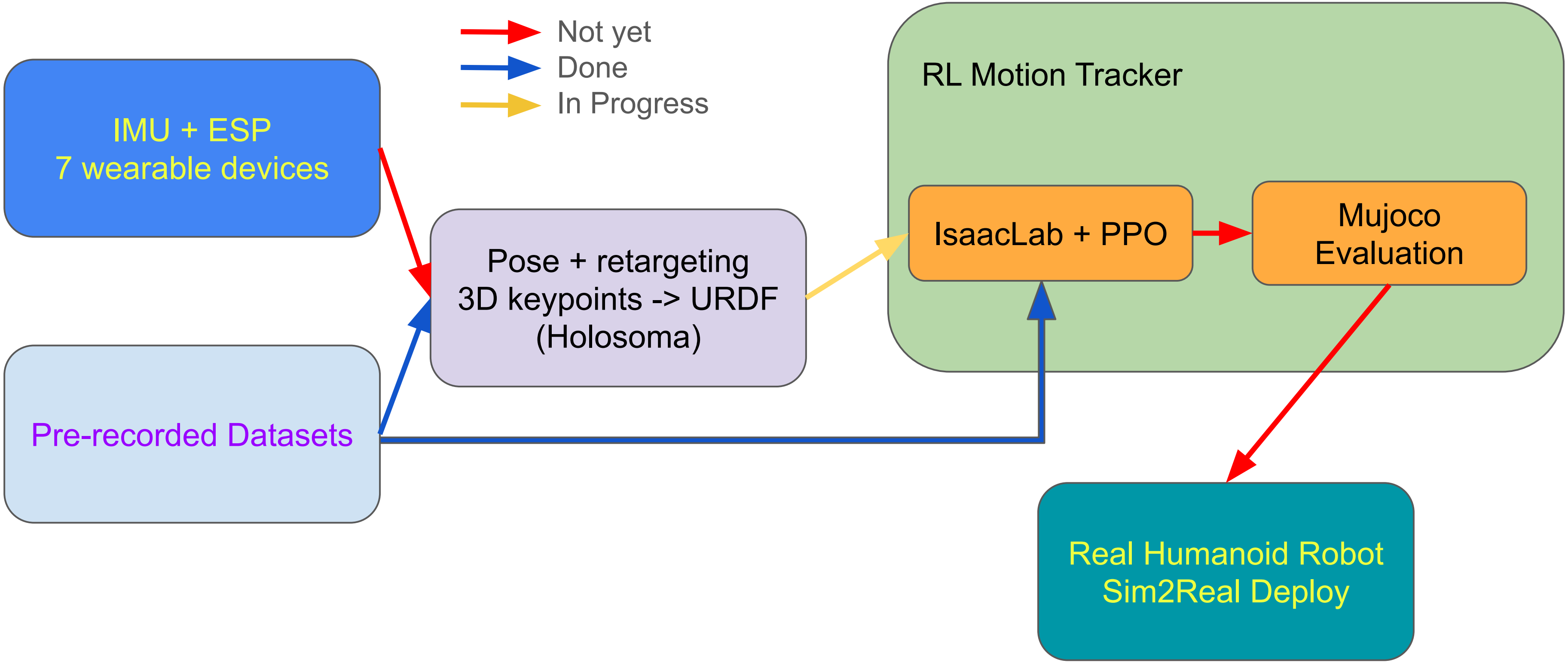


Inverse Kinematics Solver: Uses the orientation of the segments to solve for the orientation of joints and apply to the skeletal model to solve for actual pose.

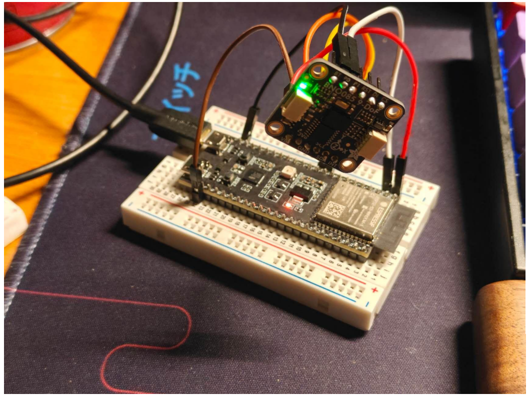
```
segment       sensor  hz     age_ms  rj_ms  sj_ms  drops  seq     quat_wxyz
pelvis        0        98.9     0.0    9.0    0.5  0      2884     0.479  0.592  0.628  0.157
left_thigh    --      --     --      --     --     --     --      --
left_shank    --      --     --      --     --     --     --      --
left_foot     --      --     --      --     --     --     --      --
right_thigh   --      --     --      --     --     --     --      --
right_shank   --      --     --      --     --     --     --      --
right_foot    --      --     --      --     --     --     --      --

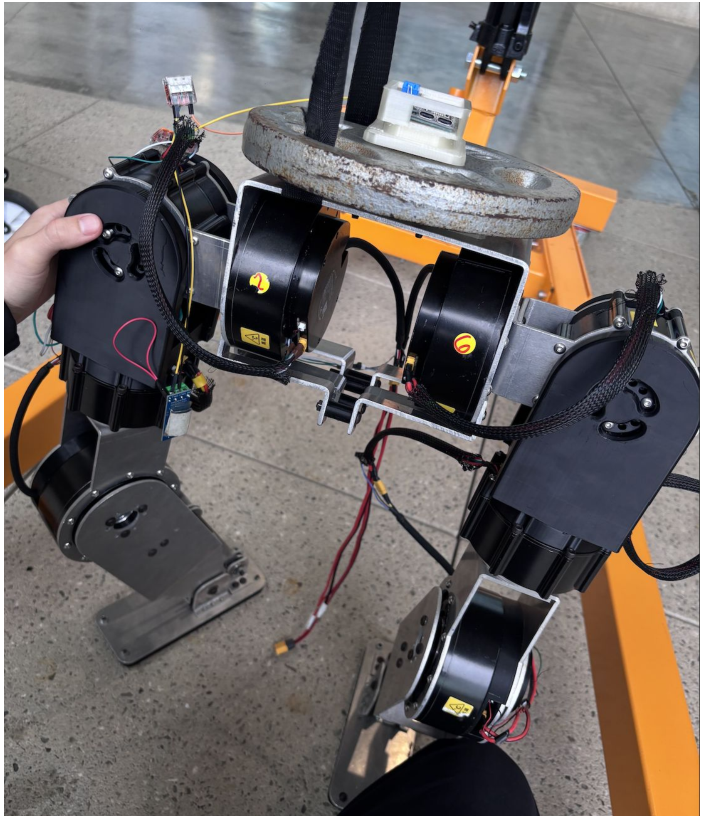
complete fresh frame: False
```

Project Objectives



MVP (Embedded)





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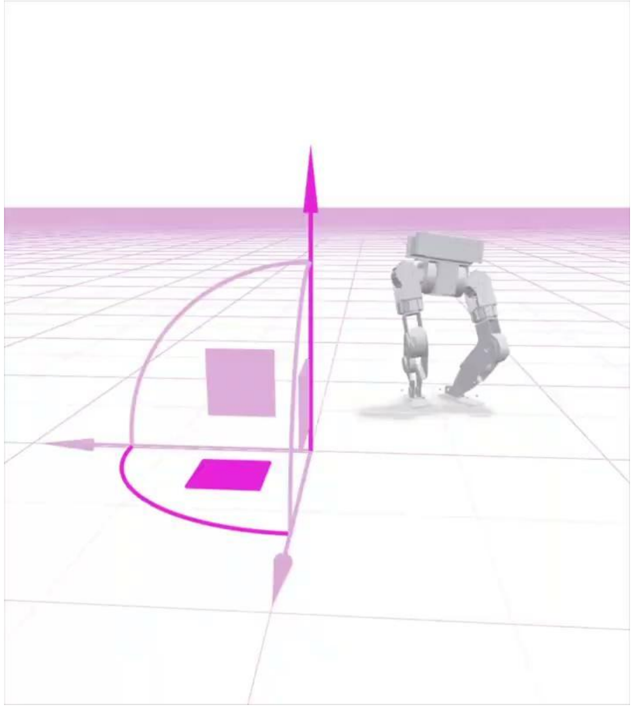
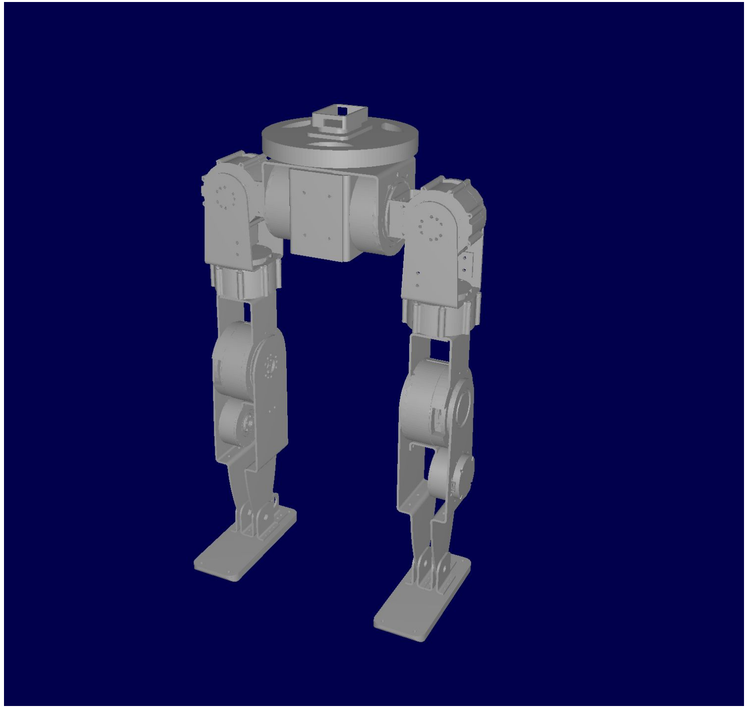
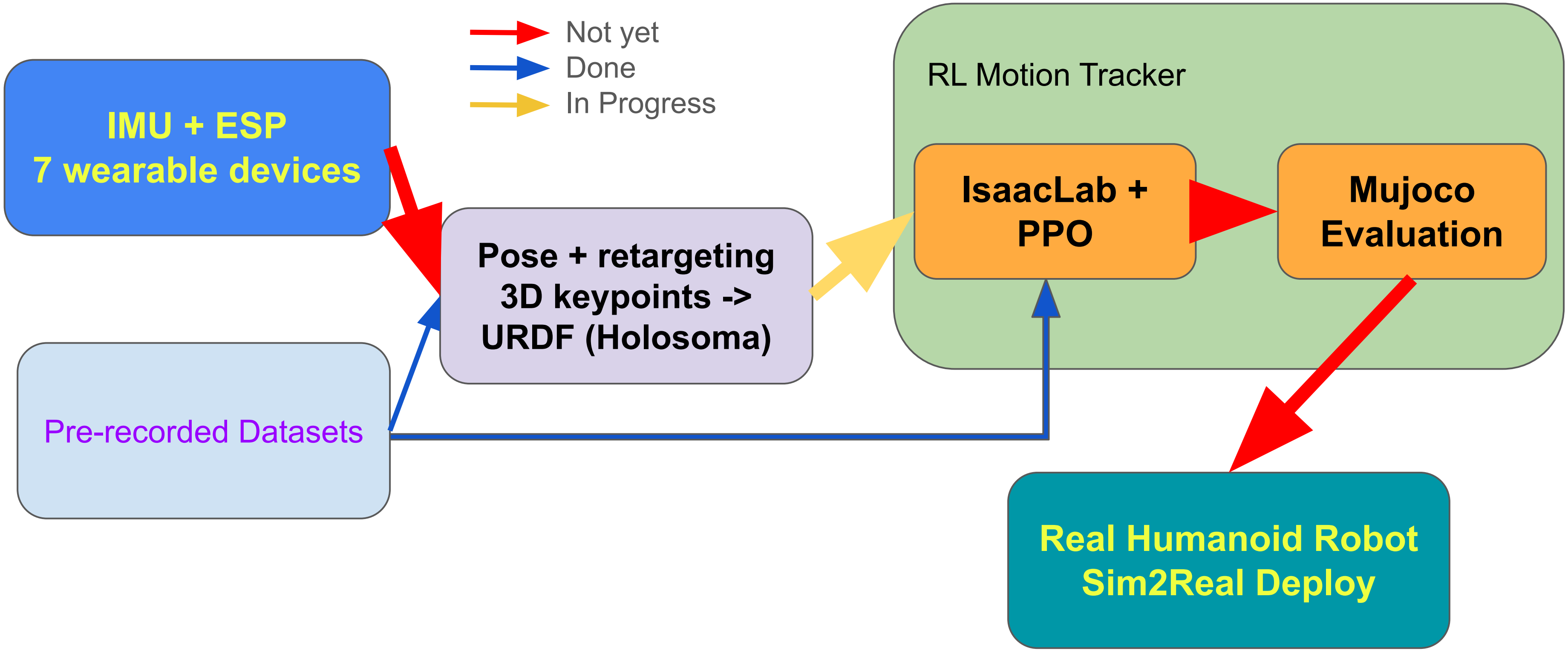
complete fresh frame: False
```

MVP (ML)





Quarter Plan



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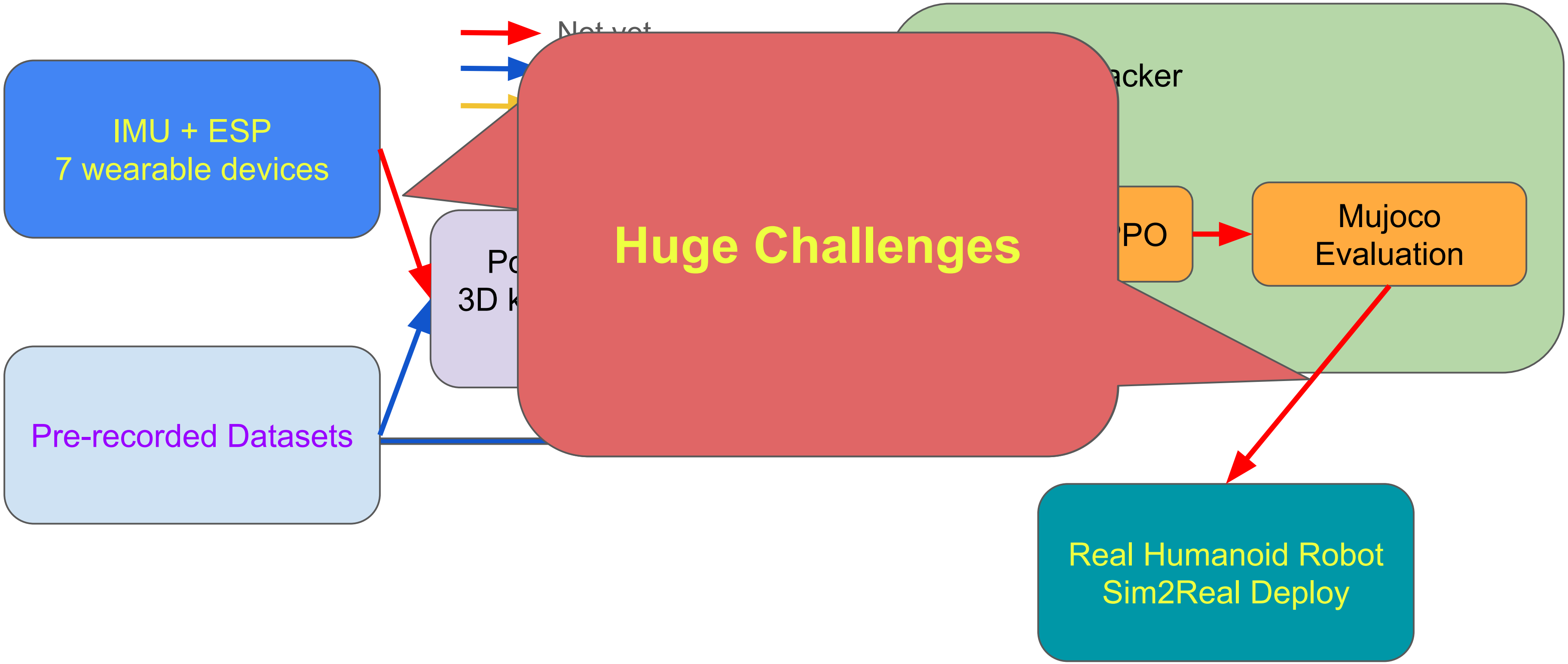


Quarter Plan

7 v

Pre-

Might switch our final demo to have embedded device controlling the humanoid robot in simulation rather than on real robot

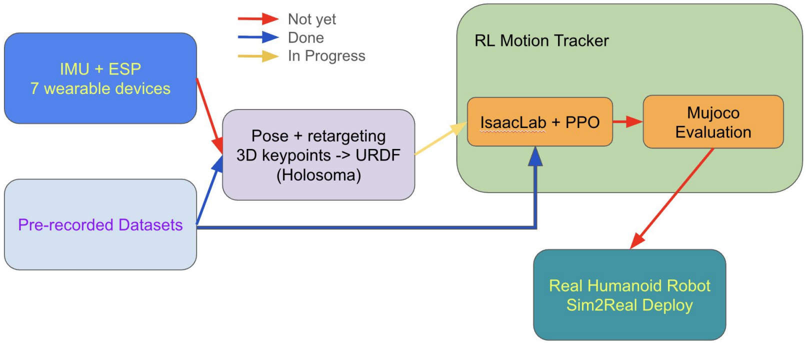
RL Motion Tracker

IsaacLab + PPO

Mujoco Evaluation

Real Humanoid Robot Sim2Real Deploy

Conclusion



What's done:

- 7 embedded devices tested for connecting signals and sending back 3D vectors
- Retargeting existing data to our humanoid robot and have it up running in the simulation
- Trained locomotion in IsaacLab simulation with reinforcement learning

What's next:

- Have the embedded devices wired up and ensure power and (wifi/bluetooth) connections
- Wire up the 3D vectors that are sent from embedded devices to simulation.
- RL training on massive online data for the robot in IsaacLab
- Do evaluation and Sim2Sim generalization from IsaacLab to Mujoco